Day 1 — Assignment

Pandas Fundamentals — abalone.csv (same dataset as today's lecture)

Work through these tasks using only what we covered today. Target time: 30–45 minutes. Use abalone.csv — the exact same dataset from today's notebook.

Section 1 — Creating & Reading Data

Task 1: Build a small DataFrame from a list of two dictionaries describing fish: the first dictionary should have keys 'species', 'weight_kg', and 'price'; the second should have only 'species' and 'weight_kg' (no price). Predict what the resulting DataFrame will look like before you run it, then check.

Task 2: Import pandas and read abalone.csv into a DataFrame.

Section 2 — First Look at Your Data

Task 3: Display the first 8 rows, then the last 8 rows.

Task 4: Check the shape and the column names.

Task 5: Run info() and describe(). Based on the output alone, answer: how many total rows are there, and what is the maximum value of the Height column?

Section 3 — Getting Your Data

Task 6: Grab the 'Length' column two ways: bracket notation and dot notation. Confirm both give the same result. Then try dot notation on the 'Shell weight' column — what happens, and why?

Task 7: Use .loc[] to select rows 20 through 30, showing only the Sex and Diameter columns.

Task 8: Now get that exact same selection using .iloc[] instead, with integer positions.

Task 9: Build a mask for rows where Diameter is greater than 0.5, and use it to filter the DataFrame. Then add a second condition — Sex equal to 'M' — to that same mask.

Task 10: Rewrite the compound filter from Task 9 using .query() instead of bracket-and-mask syntax. (Hint: you'll need to handle the space in 'Whole weight'-style column names if you reference any of them — stick to Diameter and Sex and you won't need to.)

Section 4 — Groupby & Sorting

Task 11: Group the data by Sex and find the mean Shell weight for each group.

Task 12: Sort the DataFrame by Whole weight in descending order and show the top 10 rows.

Section 5 — Creating & Dropping Columns

Task 13: Rename the Rings column to age_proxy. Then create a new column called density_proxy that holds Whole weight divided by Length, using .eval().

Task 14: Drop age_proxy properly, so the change actually sticks. Confirm it's gone. Then run fillna(-1) and dropna() on the DataFrame — even though nothing will visibly change, since there's no missing data here, this is just to practice the syntax.